

IridisX GPU Compute

Jonathon Hare & Antonia Marcu / 30-9-2026

Do this first

- Open a terminal (open `cmd` on the lab PCs)
- Use ssh to connect to iridis:
 - ssh <username>@loginx002.iridis.soton.ac.uk
 - **OR**
 - ssh <username>@loginx003.iridis.soton.ac.uk
- Enter the commands one by one at <https://tinyurl.com/iridisx>

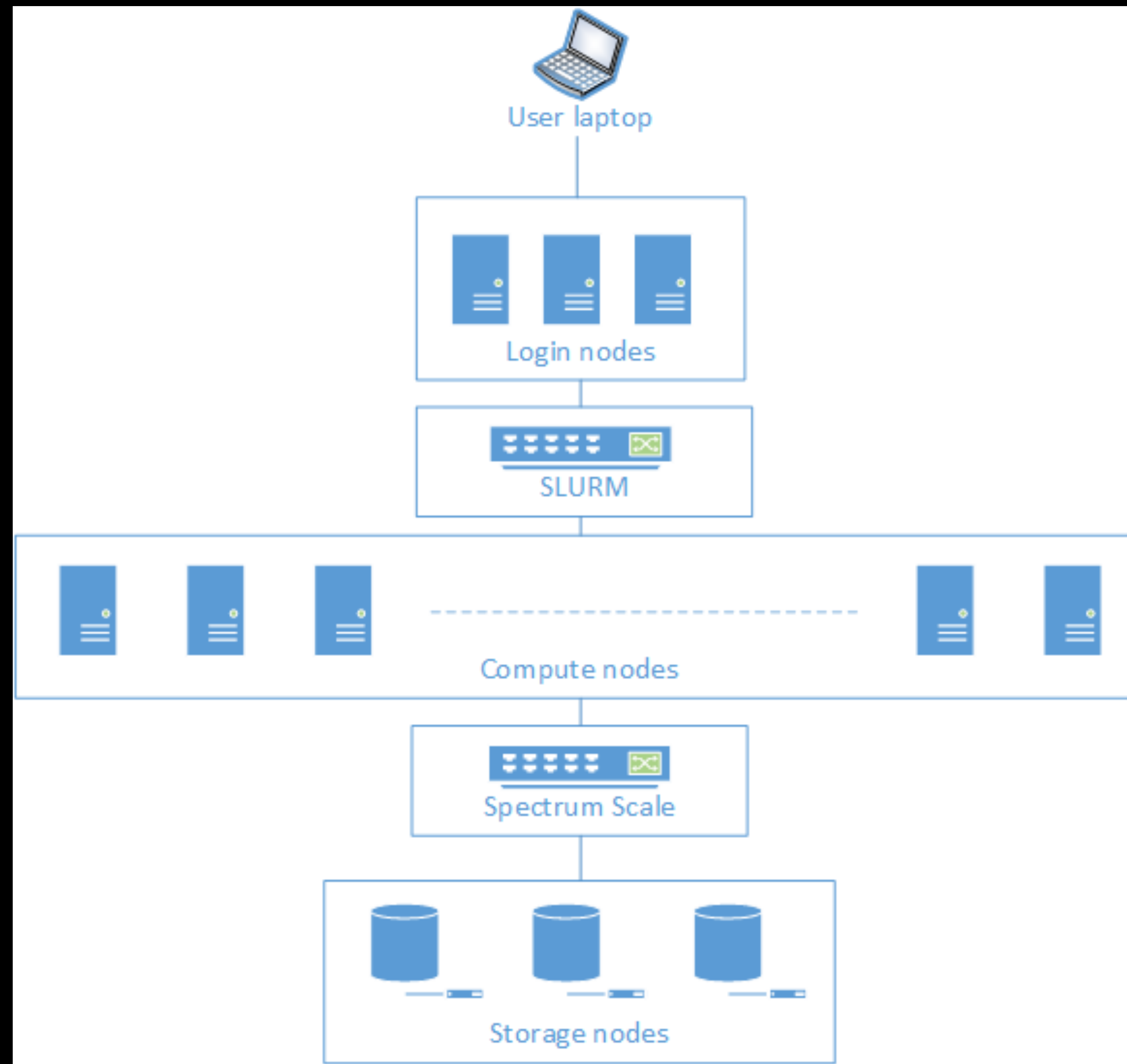
Iridis

- University supercomputer cluster
 - Interesting bit: Iridis X (Heterogeneous GPU compute: L4s to H200s, including ECS SWARM nodes with L4s, (many) quad A100s and 8-way A100s)
 - ECS Undergrads and MScs have access to the ECS Teaching L4 partition by filling in the form
 - ECS Staff and PGRs have access to all of SWARM and can request access to the full IridisX

Iridis Storage

- The various Iridis partitions have a common filesystem
- You have separate home directories and scratch directories on both and need to manually copy/sync data between them
- Many small files are your enemy... learn to use webdataset

Iridis Architecture



Iridis X Architecture

- Login nodes:
 - loginx001.iridis.soton.ac.uk (4x L4 GPUs)
 - loginx002.iridis.soton.ac.uk
 - loginx003.iridis.soton.ac.uk (4x L4 GPUs)
- /home/<username> - home directory (110GB / 204800 inodes) (backed up)
- /scratch/<username> - temp storage (1500GB / 512000 inodes)
- /iridisfs/vlcgroup/ - 80ish TB ;) - we store curated versions of common datasets here

Basics

- Compute nodes don't have internet access
 - You can run tensorboard on a login node and connect to it within the Southampton network
 - You can ssh tunnel Jupyter Notebooks (but that's maybe not the best use of Iridis)
 - You can still use WandB in *offline mode*
 - https://docs.wandb.ai/support/run_wandb_offline/
- Don't run long/intense jobs on the login node!
- Put your python envs on /scratch as they use lots of inodes
- Use myquota

Modules

- Iridis nodes use a system called Lmod to manage centrally installed software
 - `module av` # list all available modules
 - `module list` # list all loaded modules
- `module load conda/python3` # latest conda on Iridis X
- `module load gcc` # load a C compiler

Slurm

- sbatch
 - sbatch <script.sh>
- sinteractive
- sinfo
- squeue
 - squeue --me # list my jobs
- scancel
 - scancel <jobid> # cancel job <jobid>
 - scancel --me # cancel all jobs
- sacctmgr
 - sacctmgr show qos h200 format=name,MaxTRESPU%50

Slurm Partitions (Iridis X)

- a100
- a100_dev
- amd
- amd_dev
- amd_serial
- amd_student
- dual_h200
- ecsstudents_l4
- l4
- l40
- maths_a100
- mi300x
- quad_h200
- i7_h200
- scavenger_4a100
- scavenger_8h100
- scavenger_l4
- scavenger_mathsa100
- swarm_a100
- swarm_h100
- swarm_l4

sbatch

- Used to launch shell scripts which define a “job” to run on compute nodes
 - You can pass additional arguments to the script:
 - `sbatch launch.sh --config path/to/config.yaml`
 - inside the script use `$@`
- Job arrays let you launch multiple jobs with a different parameter set
 - <https://blog.ronin.cloud/slurm-job-arrays/>
 - Useful for multiple runs with different seeds, different configs, ...

Anatomy of an sbatch script

single-gpu / single node example

```
#!/bin/bash -l
#SBATCH -p ecsstudents_l4
#SBATCH --mem=24G
#SBATCH --gres=gpu:1
#SBATCH --nodes=1
#SBATCH -c 8
#SBATCH --time=1:00:00

echo "$@"

module load conda/python3
source activate feature-importance

python train.py "$@"
```

Lets go:

Hello World

- Connect to Iridis X
- Clone starter code
 - <https://github.com/ecs-vlc/iridis101/>
- Queue a simple job
 - `sbatch simple-sbatch-example.sh`

Lets go:

Training a neural network

- Download the dataset
 - `python cifar.py`
 - Ctrl-c once its downloaded - you don't want to train on the login node!
- Queue a real job
 - `launch.sh`
 - Monitor resource utilisation using `nvidia-smi` and `top`